

Reduction of the Angular Amplitude Observable: the J_3 Split as Accumulated Oriented Symplectic Area

(internal note, prior to the Weil–BFS computation)

Cosmochrony programme — working note

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Abstract

This internal note fixes the numerical observable for the absolute normalisation of the angular generation split of Q14, §6 [5], before any Weil–BFS computation is written. A short Baker–Campbell–Hausdorff reduction identifies the per-step J_{Π} -odd J_3 component of the metaplectic cascade generator with the oriented symplectic area swept per step — the central, Carnot-degree-2 increment of Heis_3 — and not with a free shear parameter. The quantity to be measured is therefore the accumulated oriented J_3 -odd area along the Weil–BFS cascade, normalised by the projected capacity $\widehat{I}(n)$; at the intrinsic saturation rank n_3^{obs} this equals $\varepsilon_{\text{Weil}}(n_3^{\text{obs}})$, to be compared with the dictionary value $\varepsilon = \frac{1}{10}$ *without fitting*. In the O12 construction the two sectors are the two terms of the fingerprint exponent: the frequency B_c carries the radial capacity σ_c , the central phase A_c carries the angular split.

Keywords: metaplectic cascade; oriented symplectic area; Heisenberg grading; generation split; Weil representation; projected capacity.

1 Context

Q14, §6 [5] establishes the amplitude relation $\varepsilon = \mathcal{N}_{\text{casc}}|\alpha|$ with a factorised architecture: the angular J_3 split (metaplectic, norm-preserving) and the radial capacity decay (projective, norm-changing) are Hilbert–Schmidt orthogonal generators coupled only by the cascade arrow, and $\mathcal{N}_{\text{casc}}$ is a pure angular–radial transfer constant. The profile and saturation status of the amplitude are fixed; its absolute value is deferred to the physical normalisation of the metaplectic J_3 generator, the shear value $\alpha \simeq 0.156$ being a diagnostic coefficient only. This note fixes the observable that a Weil–BFS computation must measure to test that absolute value; it does not attempt to prove $\varepsilon = \frac{1}{10}$.

2 Reduction lemma

Lemma 1 (Per-step split is the oriented symplectic area). *Let the cascade step be the oriented metaplectic product $g = S_X(t) S_Y(s)$ acting on the weight space $V = \mathbb{C}^2$ and lifted to $\text{Sym}^2(V)$, with X, Y the horizontal $\mathfrak{sl}(2)$ generators and $[X, Y] = 2J_3$. Then, to second Baker–Campbell–Hausdorff order,*

$$\log(S_X(t) S_Y(s)) = tX + sY + \frac{ts}{2} [X, Y] + O(3),$$

and the projection onto the J_3 direction is

$$\Pi_{J_3} \log(S_X(t) S_Y(s)) = ts J_3 + O(3).$$

Hence the per-step J_Π -odd J_3 component is

$$\alpha_{\text{step}} = ts = \text{oriented symplectic area swept per cascade step},$$

the central (Carnot-degree-2) increment of Heis_3 , and not a free shear parameter.

Proof. The second-order BCH expansion of $\log(e^{tX}e^{sY})$ gives the stated form, the only second-order term being $\frac{ts}{2}[X, Y]$ since X, Y are nilpotent of order two in the lift. With $[X, Y] = 2J_3$ this contributes $\frac{ts}{2}(2J_3) = tsJ_3$, while $tX + sY$ carries no J_3 component. The dependence on t and s only through the product ts is the oriented-area form: ts is the symplectic area of the ordered step, equal to the central-coordinate increment of the Heisenberg law $(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab')$. $\square$

Remark 1. This is the dynamical, J_Π -odd counterpart of the static result that the covariance $[1 : \frac{1}{2} : \frac{1}{2}]$ and its factor 2 originate in the Carnot-degree-2 weight of the central generator $Z = [X, Y]$ [3], in the identification with $\text{Sym}^2(V_\rho)$ of [2]. The static factor is carried by the J_Π -even Carnot dilation; the split is carried by the J_Π -odd oriented area, a distinct direction.

3 The Weil–BFS observable

Lemma 1 fixes what the computation must measure: not a chosen $\alpha(t, s)$, but the oriented central area actually accumulated by the Weil cascade on $\text{Cay}(\text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$. In the O12 construction this area is explicit. Built from the three accumulated generator steps (a_i, b_i, g_i) on the uniform seed, the fingerprint of a shell element g is a single additive character

$$\text{fp}_c(g; x) = q^{-3/2} \exp\left(\frac{2\pi i}{q}(A_c(g) + B_c(g)x)\right), \quad B_c(g) = \sum_i c_i b_i, \quad A_c(g) = \sum_i c_i (g_i - b_i a_i),$$

verified single-frequency to machine precision [6, 4]. The two terms of the exponent separate the two sectors exactly:

$$B_c \text{ (frequency)} = \text{radial} / \text{capacity}, \quad A_c \text{ (central phase)} = \text{angular} / J_3.$$

The radial capacity is the frequency-counting functional

$$\sigma_c(n) = \frac{|\{B_c(g) : g \in S_n\}|}{|S_n|},$$

blind to A_c : scaling the whole block $c \mapsto \omega c$ permutes $B_c \mapsto \omega B_c$ bijectively and leaves σ_c invariant ($[H\text{-color}]$), while $A_c \mapsto \omega A_c$ carries the per-vector phase that Gram–Schmidt discards. The angular increment is therefore the J_Π -odd, oriented shell increment of the central phase A_c ,

$$\Delta A_{J_3}^{\text{odd}}(n) = \text{Odd}_{J_\Pi} [\text{oriented shell increment of } A_c(g), g \in S_n],$$

where Odd_{J_Π} is the part odd under conjugation $c \mapsto q - c$ and the orientation is the cascade arrow. Define

$$\varepsilon_{\text{Weil}}(n) = \sum_{m \leq n} \Delta A_{J_3}^{\text{odd}}(m), \quad K_{\text{ar}}^{\text{Weil}}(n) = \frac{\varepsilon_{\text{Weil}}(n)}{\hat{I}(n)}, \quad \hat{I}(n) = \frac{\sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)}{\sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n_{\text{sat}})},$$

with $\sigma_{\text{pair}} = \sigma_c \sigma_{q-c}$. Since $\hat{I}(n_3^{\text{obs}}) = 1$ at the intrinsic saturation rank [1], the quantity to be compared with the dictionary value is

$$K_{\text{ar}}^{\text{Weil}}(n_3^{\text{obs}}) = \varepsilon_{\text{Weil}}(n_3^{\text{obs}}).$$

This eliminates the principal false positive: no shear parameters are chosen; the radial sector is the frequency count of B_c and the angular sector is the central phase A_c , separated by the form of the fingerprint itself.

4 Required outputs and honest outcomes

The Weil–BFS computation must return three quantities, not one:

1. $K_{\text{ar}}^{\text{Weil}}(q)$ at the saturation rank, for $q \in \{61, 101, 151\}$;
2. the profile $K_{\text{ar}}^{\text{Weil}}(n)$ (constant / saturating / n_3 -concentrated), cross-checking the locking law of Q14, §6;
3. the sign control under cascade reversal: the J_3 -odd component must flip sign, as required by the oriented arrow.

The comparison is $K_{\text{ar}}^{\text{Weil}}(q) \rightarrow \frac{1}{10}$, tested, not fitted. The honest outcomes are: $K_{\text{ar}}^{\text{Weil}} \rightarrow \frac{1}{10}$ is an orthogonal confirmation of ε (the ADE-deficit radial route and the metaplectic-area angular route agree); $K_{\text{ar}}^{\text{Weil}} \neq \frac{1}{10}$ but q -stable means the ADE dictionary fixes ε while the angular sector carries an additional transfer factor, with the common scale $\kappa = \frac{5}{12}$ of the deficit dictionary a natural candidate; $K_{\text{ar}}^{\text{Weil}}$ q -unstable leaves the amplitude open.

5 The deferred normalisation $A \rightarrow \varepsilon$

The fingerprint gives a concrete angle per element, $\theta_A(g) = 2\pi A_c(g)/q$. The angular amplitude is then

$$\varepsilon_{\text{Weil}} = \mathcal{N}_A \cdot \text{Odd}_{J_{\Pi}}\left(\frac{2\pi A_c}{q}\right)_{\hat{I}\text{-normalised}},$$

where $\mathcal{N}_A$ is not a fit to $\frac{1}{10}$ but the geometric factor converting a Weil central phase into the spectral coefficient of the J_3 generator on $\text{Sym}^2(V_\rho)$. It is the angular analogue of the radial scale $\kappa = \frac{5}{12}$ of the deficit dictionary. This is the last open step, and it is posed only after the observable above is fixed, so that $\mathcal{N}_A$ acts on the concrete pipeline quantity θ_A rather than on an abstract area.

References

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